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(30pts) 1. If $f(x) = \sin\left(\sin\left(\sin\left(\sin(5^{\cos x})\right)\right)\right) + \dfrac{3^{4^x+x^7}+x}{\sqrt{x^8+\tan x}} + \csc(6^x)\cdot\dfrac{8}{x^4} - x^4\sin(x)$.

cos x here ↓ , −sin x ↓ here

$\cos(\sin(\sin(\sin(5^{\cos x}))))\cdot\cos(\sin(\sin(5^{\cos x})))\cdot\cos(\sin(5^{\cos x}))\cdot\cos(5^{\cos x})\cdot(5^{-\sin(x)}\cdot\ln 5)$

move this here and instead have $(x^8+\tan x)^{1/2}$ instead

$+$

$\dfrac{(3^{4^x+x^7}+x)\left(\frac{1}{2}(8x^7+\sec^2(x))(x^8+\tan x)^{-\frac{1}{2}}\right) - (\sqrt{x^8+\tan x})(3^{4^x+x^7}\cdot\ln(3)\cdot(4^x\ln 4+7x^6)+1}{(\sqrt{x^8+\tan x})^2}$

Swap order of terms in Quotient Rule

$+$

$(-\cot(6^x)\csc(6^x)\cdot 6^x\ln(6))\left(\frac{8}{x^4}\right) + \left(\frac{-32}{x^3}\right)(\csc(6^x))$

exponent 5

$-$

$4x^3\sin(x) + \cos(x)\,x^4$

parens here

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Remove this cos (and the others with slash)

$f'(x) = \cos(\sin(\sin(\sin(5^{\cos x})))) \cdot \cos(\cos(\sin(\sin(5^{\cos x})))) \cdot 5^{\cos x} \ln(5) \cdot \sin(x)))) +$

$\cos(\cos(\cos(\sin(5^{\cos x})))) \cdot \cos(\cos(\cos(\cos(5^{\cos x}) \ln(5) \cdot \sin(x)))) +$

insert minus

$\left(3^{4^x+x^7} \ln(3) \ln(4) 4^x + 7x^6 + 1\right)\left(\sqrt{x^8+\tan(x)}\right) - \left(3^{4^x+x^7} + x\right)$

$\left(\frac{1}{2}(x^8+\tan(x))^{-1/2} \cdot 8x^7 + \sec(x)^2\right)$

$\left(\sqrt{x^4+\tan x}\right)^2$

$\csc(6^x)(-32x^{-5}) + (-\csc(6^x)\cot(6^x)\ln(6)(6^x) \cdot 8x^{-4}$

$(x^4 \cdot \cos(x)) + (4x^3 \cdot \sin x)$

Underlined texts on lines 11, 12, and 17 must be in parentheses due to order of operations

It's not that sin's turn into cos's

28/40

28/30

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perhaps forgot this was here? (Product Rule)

$f'(x) = [\cos(\sin(\sin(\sin(5^{\cos(x)})))) \cdot \cos(\sin(\sin(5^{\cos x}))) \cdot \cos(\sin(5^{-\cos x})) \cdot \cos(5^{(\cos x)}) \cdot 5^{\cos x} \cdot \ln 5 \cdot (-\sin(x))] + [(\sqrt{x^8+\tan(x)} \times (3^{4^x+x^7} \cdot \ln 3 \cdot (4^x \cdot \ln 4 + 7x^6) + 1) - (3^{4^x+x^7} \cdot \frac{8x^7+\sec^2(x)}{\sqrt{x^8+\tan(x)}}))) / (x^8+\tan(x))] + (-\csc(6^x)\cot(6^x) \cdot (6^x) \cdot 6^x \ln 6) - [x^4\cos(x) + 4x^3 \cdot \sin(x)]$

Remove this

missing 2 in front of $\sqrt{}$

28/40

28/30

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$$f'(x) = \cos(\sin(\sin(\sin(5^{\cos x})))) \cdot \cos(\sin(\sin(5^{\cos x}))) \cdot \cos(\sin(5^{\cos x})) \cdot \cos(5^{\cos x}) \cdot 5^{\cos x} \cdot \ln 5 \cdot -\sin x$$

$$+ \frac{(3^{4^x+x^7} \cdot \ln 3 \cdot (4^x \cdot \ln 4 + 7x^6) + 1)(\sqrt{x^8+\tan x}) - (3^{4^x+x^7}+x)\left(4x^3 + (\sec^2 x)\left(\frac{1}{2\sqrt{x^8+\tan x}}\right)\right)}{(\sqrt{x^8+\tan x})^2}$$

move this here

instead, put $x^8+\tan x$ here.

$$+ (-\cot(6^x)\cdot\csc(6^x)\cdot 6^x \cdot \ln 6)(8x^{-4}) + (\csc(6^x))(8(-4)x^{-5})$$

$$-(4x^3)(\sin(x)) + (x^4)(\cos(x))$$

need parens due to Order of Operations

32/40

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$f'(x) = \cos(\sin(\sin(\sin(5^{\cos x})))) \cdot \cos(\sin(\sin(\sin(5^{\cos x})))) \cdot \cos(\sin(\sin(5^{\cos x}))) \cdot$

$\cos(5^{\cos x}) \cdot 5^{\cos x} \ln 5 \cdot (-\sin(x)) +$

$\left((3^{4^x+x^7}+x) \cdot \left(\frac{1}{2}(x^8+\tan x)^{-1/2} \cdot (8x^7+\sec^2 x)\right) - (3^{4^x+x^7} \ln 3 \cdot 4^x \ln 4 + 7x^6 + 1) \cdot (\sqrt{x^8+\tan x}) \right)$

$(x^8+\tan x)$ ← Need parens ; $4^x \ln 4$ ← Need parens

$+ \csc(6^x) \cdot -32x^{-5} + \frac{8}{x^4} \cdot (-\csc(6^x) \cdot \cot(6^x) \cdot 6^x \ln 6) - x^4\cos(x) + 4x^3\sin(x)$

Swap order of terms

Need parens

36/40

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$f(x) = \sin(\sin(\sin(\sin(5^{\cos x})))) + \frac{3^{4^x+x^7}+x}{(x^8+\tan x)^{\frac{1}{2}}} + \csc(6^x)\cdot 8x^{-4} - x^4\sin(x)$

$f'(x) = \cos(\sin(\sin(\sin(5^{\cos x}))))\cdot\cos(\sin(\sin(5^{\cos x})))\cdot\cos(\sin(5^{\cos x}))\cdot\cos(5^{\cos x})\cdot 5^{\cos x}\ln 5\cdot -\sin(x)$

$+\left[(x^8+\tan x)^{\frac{1}{2}}\cdot(3^{4^x+x^7}\ln 3\cdot 4^x\ln 4 + 7x^6 + 1) - (3^{4^x+x^7}+x)\cdot\left(-\frac{1}{2}(x^8+\tan x)^{-\frac{1}{2}}\cdot 8x^7+\sec^2 x\right)\right]$

$\overline{\left((x^8+\tan x)^{\frac{1}{2}}\right)^2}$

$+\csc(6^x)(8\cdot -4x^{-5}) + (8x^{-4})(-\csc(6^x)\cot(6^x)\cdot 6^x\ln 6)$

$-(x^4\cdot\cos x) + (\sin(x)\cdot 4x^3)$

Underlined texts must be in parentheses due to Order of Operations.

37/40

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$f'(x) = \cos(\sin(\sin(\sin(5^{\cos x})))) \cdot (\cos(\sin(\sin(5^{\cos x}))) \cdot \cos(\sin(5^{\cos x}))$
$\cdot \cos(5^{\cos x}) \cdot 5^{\cos x} \ln 5 \cdot (-\sin(x)) +$

$((x^8+\tan x)^{\frac{1}{2}})(3^{4^x+x^7}\ln 3 \cdot 4^x \ln 4 + 7x^6 + 1) - (3^{4^x+x^7}+x)(\frac{1}{2}(x^8+\tan x)^{-\frac{1}{2}}$

$\cdot \dfrac{8x^7 + \sec^2 x}{(x^8+\tan(x))^{\frac{1}{2}}}) + (-32x^5 \cdot \csc(6^x) - \csc(6^x)\cot(6^x) \cdot 6^x \ln 6 \cdot \dfrac{8}{x^4})$

$-(x^4\cos(x) + 4x^3\sin(x))$

Underlined texts on lines 10 and 12 must be in paretheses due to Order of Operations

38/40

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$(\cos(\sin(\sin(\sin(5^{\cos x}))))\cdot\cos(\sin(\sin(5^{\cos x})))$
$\cdot\cos(\sin(5^{\cos x}))\cdot\cos(5^{\cos x})\cdot 5^{\cos x}\ln(5)$
$\cdot -\sin x) + ((3^{4^x+x^7}\ln(3)\cdot 4^x\ln(4)+7x^6+1)$
$(x^8+\tan x)^{\frac{1}{2}} - (3^{4^x+x^7}+x)(\frac{1}{2}(x^8+\tan(x))^{-\frac{1}{2}}$
$(8x^7+\sec^2 x))\div(x^8+\tan x)^2 + (\csc(6^x)$
$\cdot -32x^{-5} + 8x^4\cdot -\csc(6^x)\cot(6^x)\cdot 6^x\ln(6))$
$-(x^4\cdot\cos(x)+\sin(x)\cdot 4x^3)$

Underlined text must be in parentheses due to Order of Operations

39/40

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+ not times.

$\dfrac{(\sqrt{x^8+\tan x})(3^{(4^x+x^7)} \ln 3 \cdot (4^x \ln 4 \cdot 7x^6) + 1) -}{(\sqrt{x^8+\tan x})^2}$

Insert $(3^{(4^x+x^7)} + x)$

$-(\frac{1}{2}(x^8+\tan)^{-\frac{1}{2}} \cdot (8x^7+\sec^2 x)$ ← numerator cont.

$+((-\csc(6^x)\cdot\cot(6^x)\cdot 6x\ln(6)\cdot\frac{8}{x^4}) + (-32x^{-5}\cdot\csc(6^x))$

$-(4x^3)(\sin x) + (\cos(x)\cdot x^4)$

Insert parentheses.

37/40

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$+ \left(3^{4^{(x+x^7)}}\right) \ln(3) \cdot 4^{(x+x^7)} \ln(4) \cdot (1+7x^6) + \frac{1}{\left(\sqrt{x^8+\tan(x)}\right)^2} \cdot \left(-\frac{1}{2}(x^8+\tan x)^{-1/2} \cdot x^8 \ln(8) + \frac{1}{2}\tan(x)^{-1/2} \cdot \sec^2(x)\right) \cdot$

$+ \csc(6^x) \cdot \cot(-4x^{-5}) \cdot -(-\csc^2(6^x) \cdot 6^x \ln(6)) \left(\frac{8}{x^4}\right) - x^4 \cdot \cos(x) + \sin(x) \cdot 4x^3$

4^{x+x^7} is the exponent, not $4(x+x^7)$

+ here because — is already here

Square root can't distribute

Must put both in parentheses

27/40